# From One Table to Many Tables

Lecture Notes · Game DB Junior

**In this lesson**

- Why **one table per database** is rare in the real world
- What comes next: a database for a **photo sharing app** with **many tables**

## 1. What we did so far

We spent a good amount of time writing basic SQL. But to be honest, this SQL did not really show a **real world situation**.

- We worked with **single tables** inside **single databases**.
- The **cities** table was the only table in its database.
- In the exercises, we worked with a totally separate database. It had one table: **phones**.

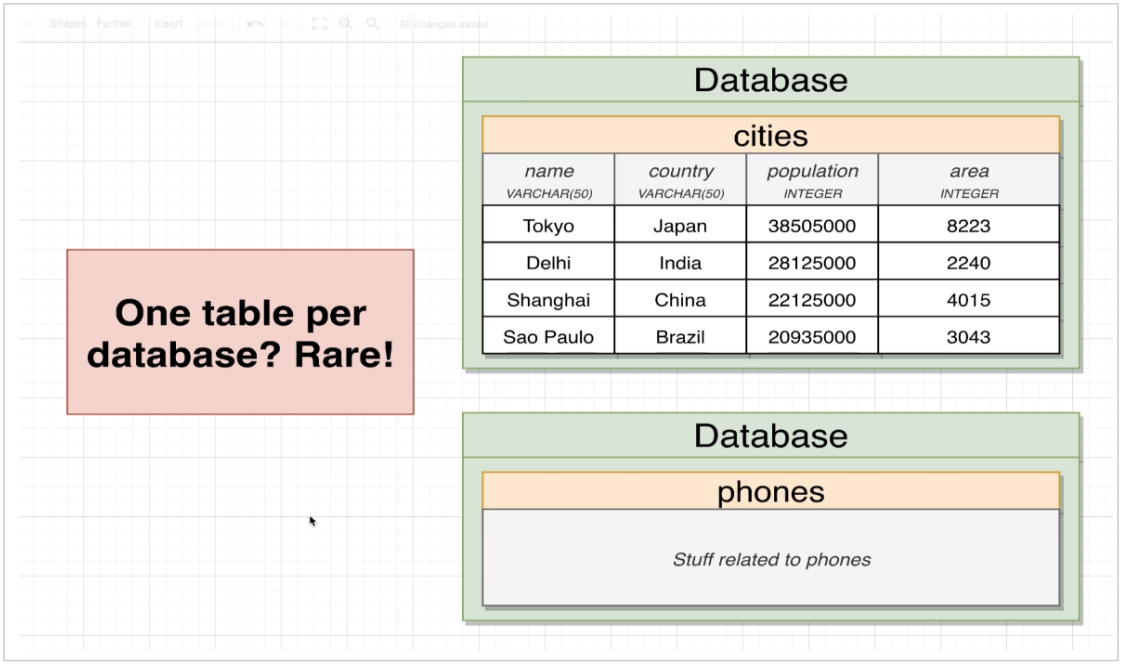



*Figure 1 — One table per database? Rare!*

## 2. One table per database? Rare!

In a real world application, it is **very rare** to have just one table of data in a database. It is possible, and it probably exists somewhere, but you will not see it often.

Working with one table would be easy and simple for learning SQL. But it does not show reality. So we make things a bit more challenging.

## 3. Next: a photo sharing app

We will start working on a database for a **photo sharing application**. We will work with **many tables** inside **one database**.

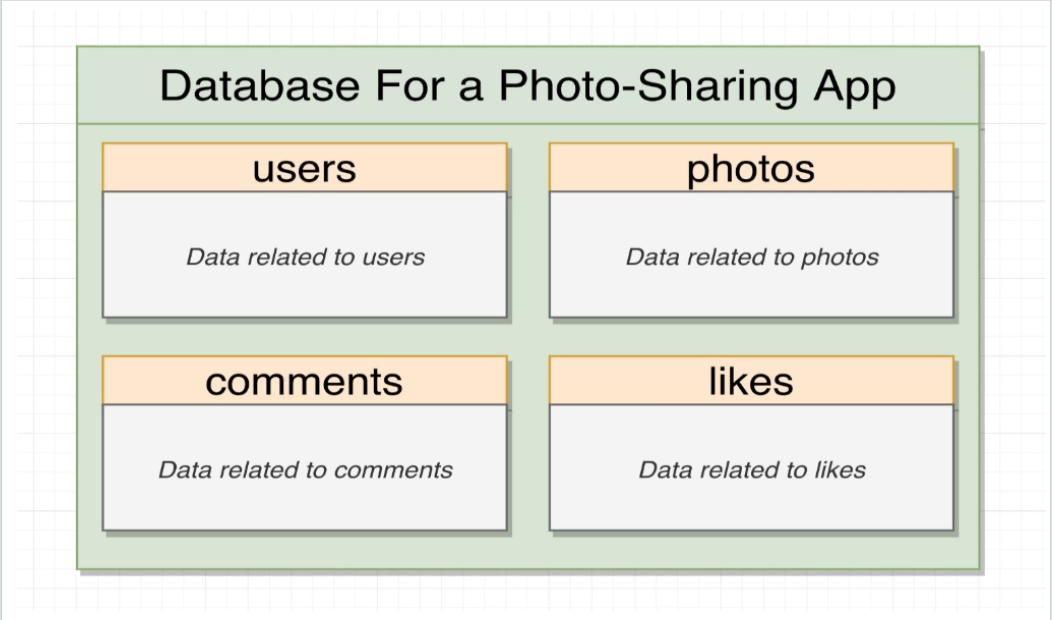



*Figure 2 — Database for a photo-sharing app*

| Table | What it holds |
|---|---|
| users | Data related to users |
| photos | Data related to photos |
| comments | Data related to comments |
| likes | Data related to likes |

**Why move to harder things so quickly?** Because it shows reality. Working with many tables is what you will really do every day when you work with Postgres. It will be a bit harder to understand, but it is a better use of our time.

## 4. Summary

| Idea | Remember |
|---|---|
| So far | One table in one database (cities, phones) |
| Real world | Having only one table in a database is rare |
| Next | One database with many tables: users, photos, comments, likes |

**Next lesson:** how we might design the photo sharing app database and which tables it needs.

**Sources**
Lecture transcript and lecture images (Figures 1–2)